Wildfire level-set PINN, status for 8 October

Written 5 October 2026, early Monday. Deadline is Thursday 8 October. Numbers below are from the trained checkpoints, not from a plan.

What you can send, and what your partners asked for

You can send a map of the 2019 Kincade Fire and the January 2025 Palisades Fire. Each file is a WGS84 GeoJSON of observed hulls and predicted lines. The checkpoint does not take a new table of their observations and return a front. Their points never enter the network at prediction time.

2 / 20
Kincade times with a predicted line

null
Scored front error, both fires

12 / 4 / 4
Kincade train, validation, test

Thu 8 Oct
Handoff day

Where the project stands

The Colab notebook is now a local package. One command downloads a named fire, builds perimeters, trains a level-set network, and writes *fronts.geojson*. Palisades is still the default. Kincade is a second dataset: Sonoma County, ignition 23 October 2019 through containment 6 November, box covering John Kincade Road and the run toward Healdsburg and Windsor. Usable satellite hulls run from 24 October 09:05 through 30 October 10:33. That is 20 overpasses, at the low end of the 20–30 target. Later days in the containment window did not produce more hulls.

Training supervision is a signed distance on every training perimeter: negative inside the observed hull, zero on the boundary, positive outside. The PDE residual is unchanged. Predictions are not shifted after inference. Every 500 epochs the log records min(φ), max(φ), the fraction of cells with φ < 0, and whether a zero contour exists. A run that stays positive on every overpass is marked degenerate. Kincade's final flag is false, because the first two overpasses do cross zero.

The two trained fires

FIRE	OVERPASSES	SPLIT	PREDICTED LINES	SCORED ERROR
Palisades, 7–12 Jan 2025	14	First 10 train, last 4 held out	Ignition overpass only	null on all 4 held-out times
Kincade, 24–30 Oct 2019	20	12 train, 4 validation, 4 test, in time order	24 Oct 09:05 and 10:46 only	null on all 8 scored times

Source: outputs/palisades/metrics.json and outputs/kincade/phi_sign.json after 4,000-epoch training on CPU. Scored error is the mean distance from the longest φ = 0 line to the observed hull. No line means the error stays null.

Kincade minimum φ after 4,000 epochs

Domain length L is 53.6 km. Interior targets on the early hulls are only a few kilometers deep, so a field that sits about +2 to +3 km everywhere can still keep the loss small. From the evening of 24 October

onward, the minimum is positive and there is no zero contour. Validation and test were not in the loss.

Times with a φ = 0 line — 2 of 20 overpasses

INDEX	TIME	SPLIT	MIN Φ (M)	ZERO CONTOUR
0	2019-10-24 09:05	train	−430	yes
1	2019-10-24 10:46	train	−5	yes
2	2019-10-24 20:27	train	+1,998	no
3	2019-10-24 22:08	train	+2,173	no
4	2019-10-25 10:27	train	+2,426	no
5	2019-10-25 20:08	train	+2,419	no
6	2019-10-25 21:49	train	+2,447	no
7	2019-10-26 10:08	train	+2,762	no
8	2019-10-26 21:30	train	+2,900	no
9	2019-10-27 09:49	train	+2,868	no
10	2019-10-27 21:11	train	+2,888	no
11	2019-10-28 09:30	train	+2,990	no
12	2019-10-28 11:11	validation	+3,020	no
13	2019-10-28 20:52	validation	+3,231	no
14	2019-10-29 09:11	validation	+3,485	no
15	2019-10-29 10:52	validation	+3,490	no
16	2019-10-29 20:33	test	+3,446	no
17	2019-10-29 22:14	test	+3,427	no
18	2019-10-30 08:52	test	+3,253	no
19	2019-10-30 10:33	test	+3,217	no

Source: outputs/kincade/phi_sign.json. Values are the network output in meters. Nothing was added to φ after training.

Finished since the notebook

PIECE	STATE
Single entry point, main.py, and the wildfire_pinn package	Done
Palisades pipeline, 4,000-epoch checkpoint, GeoJSON	Done
Signed-distance targets inside, on, and outside each training hull	Done
Kincade as a named dataset, chronological split, 20 hulls	Done
2019 LANDFIRE terrain and fuel, 2019 gridMET, spread-rate file	Done

PIECE	STATE
Six Kincade fuel codes added with published Mx and spread class	Done
φ sign log every 500 epochs, plus a degenerate flag	Done
Evaluate, diagnostics, and playback load the checkpoint without training	Done
firms_key.txt for the NASA key, gitignored	Done
fronts.geojson with timestamps and train / validation / test labels	Done

The six fuel codes that stopped the Kincade run are 143 SH3, 146 SH6, 162 TU2, 187 TL7, 201 SB1, and 202 SB2. TU2 is the one that covers real ground, about 15,000 cells. Each code uses Scott and Burgan's extinction moisture and the existing rate tiers: high 1.0, moderate 0.2, low 0.07. The spread-rate formula and the Palisades fuel rows were left as they were.

Still open before Thursday

The scientific result is already in the files. The remaining work is the handoff and the write-up. Another 4,000 epochs of this same loss is not on the list: both fires already ran that long, and later times stayed positive.

DAY	FINISH THIS	DONE WHEN
Mon 5 Oct	One-page partner note: which file to open, what a null geometry means, and the evaluate command that does not retrain.	A teammate can read the note and open fronts.geojson without asking how.
Tue 6 Oct	Course write-up of the method and this result: signed distance, chronological split, fuel-code table, and the min-φ table above.	The null scored error is stated as the result, with the reason φ stayed positive.
Wed 7 Oct	Dry run from their side. Copy outputs/kincade/fronts.geojson and levelset_pinn.pt. A git clone does not contain them; data/ and outputs/ are gitignored.	Their map shows 20 observed hulls and 2 predicted lines, and they can say which times those lines are.
Thu 8 Oct	Send the GeoJSON, the checkpoint, and the note. Palisades can go in the same packet as the second case.	The packet matches the files on disk. It does not claim a forecast from data they collect later.

Kincade packet

- outputs/kincade/fronts.geojson — 40 features, observed hull and predicted line for each of 20 times.
- outputs/kincade/levelset_pinn.pt — weights for this fire only. Reload with *python main.py --name kincade evaluate*.
- outputs/kincade/phi_sign.json — min, max, negative fraction, and contour flag. degenerate is false.
- outputs/kincade/heldout_fronts.png — scored times. The predicted line is absent because φ never crosses zero there.

Leave out of the Thursday claim

- A model that forecasts a new fire from a perimeter they just measured. The network input is normalized time and position for one fire. Weather, fuel, and the hulls enter only while that fire is

training. Time runs from the first perimeter in the file to the last. There is no call that means “draw this 12 hours ahead.”

- A fix that subtracts min(φ) so a zero contour appears. That would invent a front. The GeoJSON already keeps times with no crossing, and it stores null geometry on purpose.
- Treating more epochs as the remaining task. At epoch 4,000 the Kincade loss is about 0.115, and minimum φ on the scored times is still about +3.0 to +3.5 km.

Kincade training finished 5 October 2026 at 00:20 local time, 4,000 epochs, about 18 minutes on CPU. Split is chronological. Palisades held-out metrics were already null after its 4,000-epoch run.